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Precision Levelling Twin Niches

28.04.2026 | Technical Discussion CD-A

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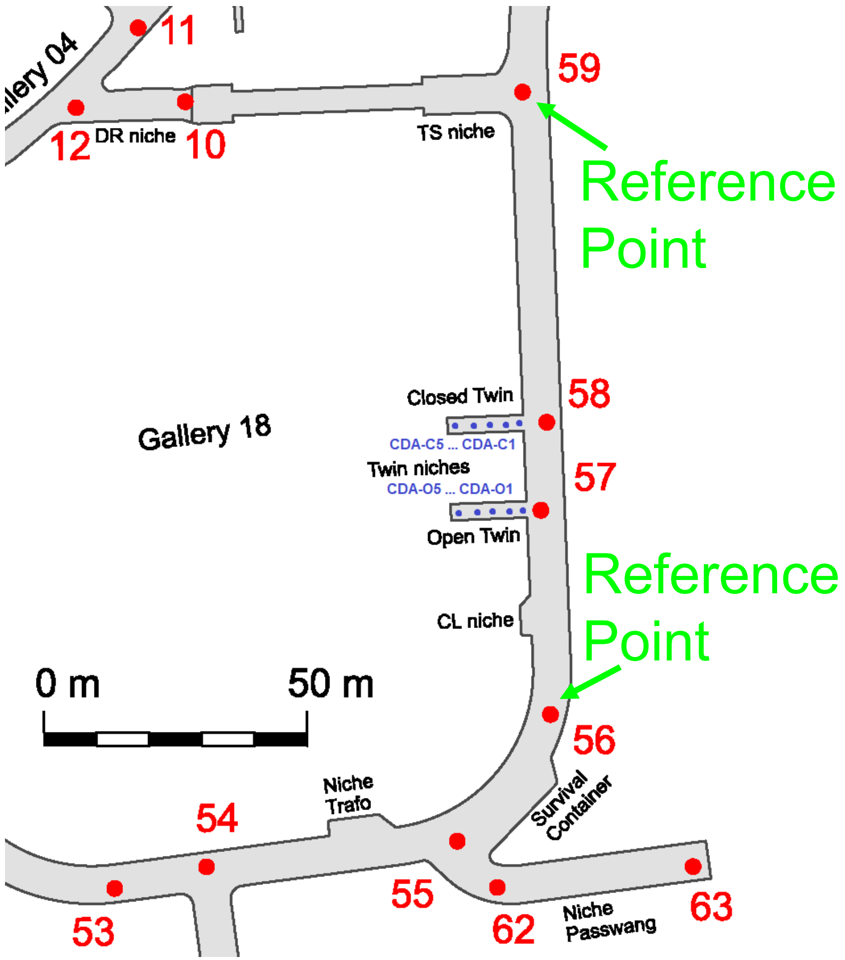
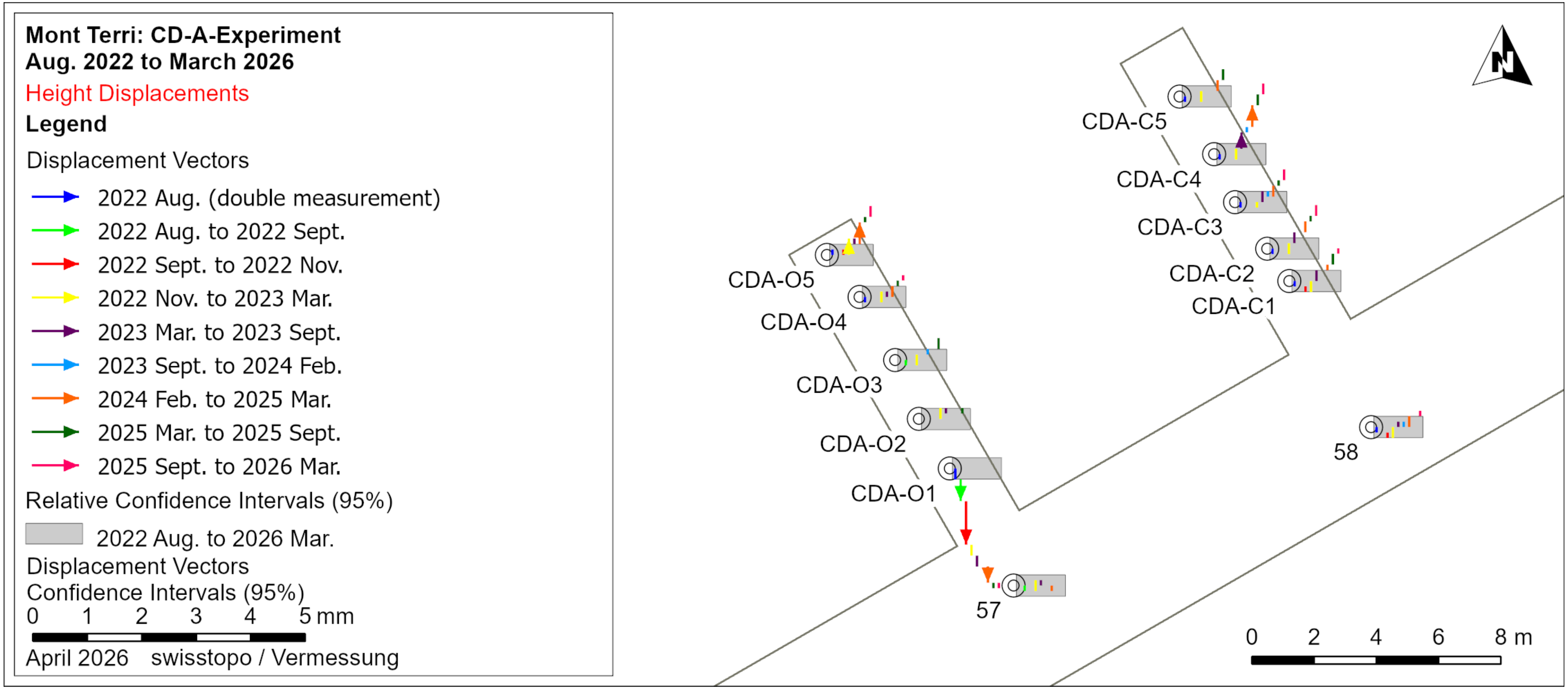
Precision Levelling Twin Niches
Structure geodetic network

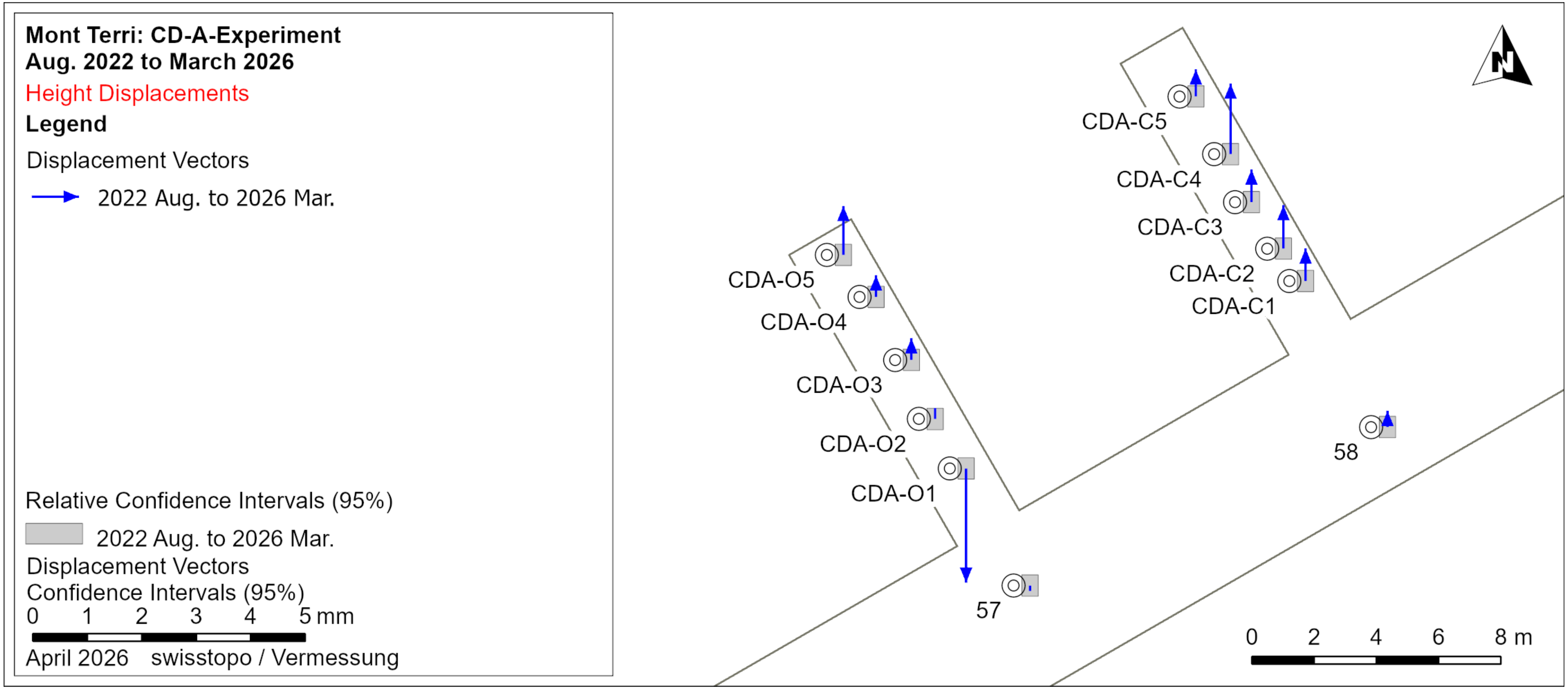
- First measurement 29/30 August 2022
- 5 points in both niches
- Reference points in the Lab
- Synchronous with the laser scanning and convergence measurements
- Relative height changes in the twin niches detectable
- Previous measurement campaigns:
  - 2022 Aug. (double measurement)
  - 2022 Sept.
  - 2022 Nov.
  - 2023 Mar.
  - 2023 Sept.
  - 2024 Feb.
  - 2025 Mar.
  - 2025 Sept.
  - 2026 Mar.



Precision Levelling Twin Niches
Current results → all intermediate epochs



Precision Levelling Twin Niches
Current results → total displacement



Precision Levelling Twin Niches
Current results → total displacement

Point name	Location	Height difference from initial measurement [mm]
57	Galerie 18	-0.1
58	Galerie 18	0.3
CDA-C1	Closed Twin	0.6
CDA-C2	Closed Twin	0.8
CDA-C3	Closed Twin	0.6
CDA-C4	Closed Twin	1.3
CDA-C5	Closed Twin	0.5
CDA-O1	Open Twin	-2.1
CDA-O2	Open Twin	0.2
CDA-O3	Open Twin	0.4
CDA-O4	Open Twin	0.4
CDA-O5	Open Twin	0.9

Precision Levelling Twin Niches Summary

- Detectable displacement: from 0.1 to 0.2 mm
  (depending on the point; Confidence level 95%)
- Largest deformations:
  - CDA-O1: -2.1 mm (settlement)
  - CDA-C4: +1.3 mm (heave / uplift)
- On all other points:
  - With the exception of CDA-O1, there is a tendency towards uplift at all points in the Twin Niches.
  - The height changes in the Open Twin are smaller (except at the aforementioned point O1).
- Outlook
  - Measurements over a longer time period
  - Direct comparison with laser scanning measurements